

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds 4 or more correct row minima/col maxima	AO1.1a	M1	Row minima = (2, 0, -2) Col maxima = (2, 3, 5)
	Correctly finds all row minima/col maxima	AO1.1b	A1	Max(row minima) = 2 Min(col maxima) = 2
	Explains correctly that as the max(row minima) is equal to the min(col maxima) then a stable solution exists and the value of the game is 2	AO2.4	E1	As Max(row minima) = 2 = Min(col maxima), then a stable solution exists and the value of the game is 2
	Allow 'Maximin' and 'Minimax'			
	Marking Instructions	AO	Marks	Typical Solution
ALT (a)	Uses dominance to reduce the size of the pay-off matrix to 2×3 or 3×2	AO1.1a	M1	A_1 (or A_2) dominates A_3 , so remove A_3 [from pay-off matrix]
	Uses dominance to reduce the pay-off matrix to 1×1	AO1.1b	A1	S_1 [now] dominates S_2 and S_3 , so remove S_2 and S_3
	OR Correctly finds all remaining row minima/col maxima			A_1 [now] dominates A_2 , so remove A_2
	Explains that each player will play the same strategy each time which results in Alex gaining 2 each game OR Explains correctly that as the max(row minima) is equal to the min(col maxima) then a stable solution exists and the value of the game is 2 Allow 'Maximin' and 'Minimax'	AO2.4	E1	Alex will only ever play A_1 and Sam will only ever play S_1 , which results in Alex gaining 2 each game. Therefore, the value of the game is 2.
(b)	Identifies strategy for each player Condone stating ' A_1 ' and ' S_1 '	AO1.1b	B1	Alex strategy A_1 Sam strategy S_1
				Total 4 marks

Q2.

(a) Min

$$\begin{pmatrix} 4 & -1 & 2 & 3 \\ 4 & 6 & 3 & 7 \\ 1 & 3 & -2 & 4 \end{pmatrix} \begin{matrix} -1 \\ 3 \\ -2 \end{matrix}$$

Max 4 6 3 7

Maximin (row) = 3

Either correct, including correct values

M1

Minimax (col) = 3

Both correct, written as equations PI by next line

A1

CSO

As Maximin (row) = Minimax (col)

There is a stable solution

Must have equation and statement and scored first 2 marks

E1

(Play safe) (H) B
 (Play safe) (W) F

Both correct

B1

4

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(b) Saddle point (B, F)

B1

1

[5]

Q3.

(a) (i)

Row min - 6, - 3, - 5, - 4

Max (row min) = - 3

Col max 5, 4, - 3

Min (col max) = - 3

*attempt to find maximin and minimax
condone one slip in values*

M1

all rows min and col max **values correct**
and max (row min) = - 3 **identified**
and min (col max) = - 3 **identified**

A1

$$\max(\text{row min}) = \min(\text{col max}) = -3$$

⇒ game has a stable solution

full statement involving maximin and minimax **and** both
values = - 3

E1

3

(ii) Adam plays **A₂** & Bill plays **B₃**

B1

1

(iii) Value of game for Bill is +3

B1

1

*Examiners must use the correct symbol for marks carried
forward, ie ringed totals with arrows through them.*

(b) (i) (Never play) **C₂**
correct strategy stated

C₂ dominated **by** C₁ (-3 > -4 and 2 > 1)

and correct reason

condone 3 < 4 and -2 < -1

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B1

1

(ii) C₁ : $3p - 2(1-p)$
either correct unsimplified

M1

$$C_3 : -3p + 5(1-p)$$

both correct unsimplified

$$\{5p - 2, 5 - 8p\}$$

A1

2

(iii) $3p - 2(1-p) = -3p + 5(1-p)$
equating their 2 expressions

M1

$$\Rightarrow p = \frac{7}{13}$$

A1

2

(iv) Value of game $= 5 \times \frac{7}{13} - 2$
or $5 - 8 \times \frac{7}{13}$

B1

1

[11]

Q4.

(a)

Row min
-2
-3
-5

Col max
4
0
-2

Row minima and column maxima (all values)

B1

$$\text{Max (row min)} = -2$$

$$\text{Min (col max)} = -2$$

Both attempted or stated/indicated

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Since these are equal, there is a stable solution

Must have both values = -2 plus statement (withhold if max (min) and min (max) not stated)

A1

Ann plays A_1 and Bill plays B_3 for play-safe

E1

4

(b) (i) Let Russ play R_1 with probability p
 And R_2 with probability $1 - p$

$$C_1 : \text{expected gain } -4p + 2(1 - p)$$

$$C_2 : 7p - (1 - p) = 8p - 1$$

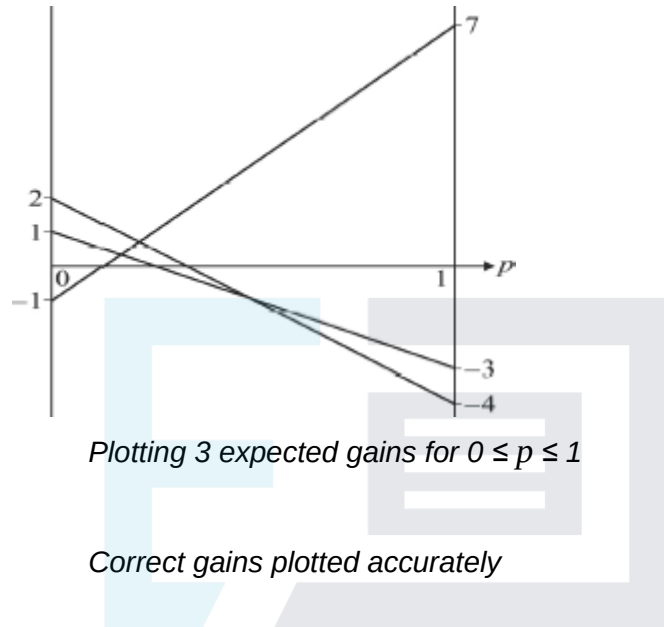
2 correct unsimplified

M1

$$C_3 : 3p + (1 - p) = 1 - 4p$$

All correct

A1



M1

A1

$$\text{Solving } 8p - 1 = 1 - 4p$$

Choosing highest point of their region or correct

M1

$$\Rightarrow p = \frac{1}{6}$$

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A1

$$\Rightarrow \text{Russ plays } R_1 \text{ with probability } \frac{1}{6}$$

$$\text{and } R_2 \text{ with prob } \frac{5}{6}$$

E1

7

(ii) Value of game = $\frac{8}{6} - 1$

Or $1 - \frac{4}{6}$

$$= \frac{1}{3}$$

B1

1

[12]

Q5.

(a)

Row minima

–7

–1

–3

Column max 6 8 –1

Max (row min) = –1

*Attempting Row Min & Col Max
or Maxmin and Minmax*

M1

Min (col max) = –1

*All values correct and shown with
correct words*

A1

Since these values are equal the game has
a stable solution

Must both be –1 and have statement

E1

Raj plays II, Cal plays Z

B1

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4

(b) (i) $C_1 : 5p - 2(1 - p)$
 $7p - 2$

B1

$$C_2 : xp + 4(1 - p)$$

B1

2

(ii) Value of game = $\frac{8}{3}$
 $\Rightarrow 5p - 2(1 - p) = \frac{8}{3}$

Their expected gain (C_1) = $\frac{8}{3}$

M1

$$\Rightarrow p = \frac{2}{3}$$

A1

$$xp + 4(1 - p) = \frac{8}{3}$$

$$\text{Their } C_2 \text{ gain} = \frac{8}{3} \text{ (must involve } x)$$

M1

$$\Rightarrow \frac{2}{3}x + \frac{4}{3} = \frac{8}{3}$$

$$\Rightarrow x = 2$$

A1

4

[10]

Q6.

- (a) Rob's gain = Con's loss
(at each entry of matrix)

Zero-sum explained

*Rob's winnings + Con's winnings = 0
(for every pair of strategies)*

E1

1

- (b)

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	-2	5	3	min -2
	3	-3	-1	-3
	-3	3	2	-3
max	3	5	3	

*min of rows and max of columns
All values correct (seen)*

B1

$$\left. \begin{array}{l} \text{maximin} = -2 \\ \text{minimax} = 3 \end{array} \right\} \text{either correct}$$

M1

$$\begin{array}{l} -2 \neq 3 \\ \Rightarrow \text{no stable solution} \end{array}$$

E1

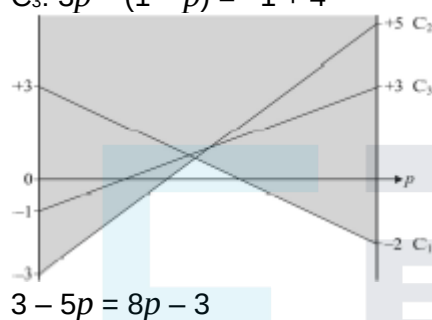
3

- (c) R_3 dominated by R_1
 $(-3, 3, 2) < (-2, 5, 3)$ so never play R_3

E1

1

- (d) (i) Choose R_1 with probability p
 and R_2 with probability $1 - p$
 Expected gain when C plays:
 $C_1: -2p + 3(1 - p) = 3 - 5p$
 $C_2: 5p - 3(1 - p) = 8p - 3$
 $C_3: 3p - (1 - p) = -1 + 4$



Attempt at one expression

All correct unsimplified

M1

A1

Plotting expected gain for $0 \leq p \leq 1$

M1

Correct with values at $p = 0$ and $p = 1$ clear

A1

Choosing C_1 and C_2 intersection or their highest point

M1

$$\Rightarrow p = \frac{6}{13}$$

A1

Play R_1 with probability $\frac{6}{13}$

and R_2 with probability $\frac{7}{13}$

FT their p (statement needed)

E1ft

(ii) Value of game $= 3 - \frac{30}{13}$
 $\frac{48}{13} - 3$
 $= \frac{9}{13}$

B1

1

[13]

Q7.

(a) (i)

				Row min
				-4
				-2
				-1
Col max	5	-1	3	

Attempt at row minimum and column maximum

all figures correct

M1

A1

min (col max) = max (row min)
 $\Rightarrow$ stable solution

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E1

3

(ii) Ros plays III and Col plays Y

B1

value of game = -1

B1

2

(b) (i) Ros plays R_1 with probability p and R_2 with probability $1 - p$

Expected gains when Col plays:

$$C_1 : 3p - 2(1 - p) = 5p - 2$$

$$C_2 : 2p - (1 - p) = 3p - 1$$

$$C_3 : p + 2(1 - p) = 2 - p$$

attempt at least 2

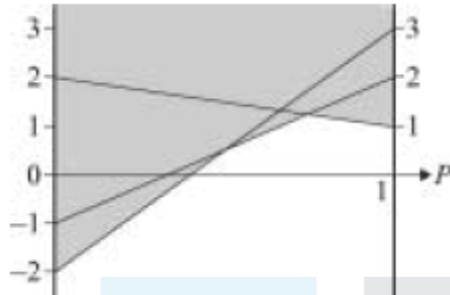
M1

correct unsimplified

A1

Plot expected gains against p for $0 \leq p \leq 1$

M1



correct (must see 0 or 1 on P axis, or implied by their numbers)
A0 if not possible to see highest point of region being correct

A1

Choose highest point of region below lines $\Rightarrow 3p - 1 = 2 - p$

must be this pair of lines or their highest point

M1

leading to $p = \frac{3}{4}$

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A1

Therefore Ros plays R_1 with prob $\frac{3}{4}$

and plays R_2 with prob $\frac{1}{4}$

ft their p from any lines

B1ft

7

(ii) Value of game $= 3 \times \frac{3}{4} - 1$
or $\left(2 - \frac{3}{4}\right) = 1\frac{1}{4}$

B1

1

Q8.

- (a) (i) Row min $(-2, 2, -3)$

M1

Max 2
 Column max $(3, 5, 2, 6)$
 Min 2

A1

$2 = 2$, therefore stable solution

E1

3

- (ii) 1 saddle point at $(2, 3)$

B1

1

- (b) Colin plays I with prob p , II with $(1 - p)$

M1

Return $p(x + 2) + 3(1 - p)$

A1

$p(x - 1) + 5(1 - p)$

A1

$$p(x - 1) + 3 = \frac{19}{5}$$

equating to value

M1

$$p(x - 6) + 5 = \frac{19}{5}$$

$$p(x - 1) = \frac{4}{5}$$

or solving for p first

A1

$$p(x - 6) = -\frac{6}{5}$$

A1

$$\frac{4}{5(x-1)} = \frac{-6}{5(x-6)}$$

M1

$$4x - 24 = -6x + 6$$

$$x = 3$$

CAO

A1

8

[12]

Q9.

(a) 7

B1

1

(b) Col. Max 8, 13, 8, 11

Min = 8

M1

A1

Row Min -7, -6, 8, 8
Max = 8

A1

3

∴ Stable solution

(c) Points (3,1) (4,1) (3,3) (4,3)

B2,1

2

[6]

Examiner reports

Q1.

Nearly 90% of students scored at least one mark in part (a). Of these students, many scored 2 marks but failed to score the third mark due to a lack of depth in their explanation of why the value of the game was 2. Many did not refer to, or were not clear enough about, the $\max(\text{row minima})$ and $\min(\text{col maxima})$ both being equal to 2 and therefore being equal to each other, and many did not then relate this back to the value of the game.

Interesting correct solutions included the use of dominance to reduce the size of the pay-off matrix to 1×1 . Of the errors that students made, the most prevalent was determining the row maxima and column minima, which lead to an incorrect conclusion and often impacted the answer to (b).

Q2.

Although the majority of students had an idea of the requirements of this question, full marks were rarely achieved. It is essential that students give full and clear statements to provide evidence to the examiner; simply writing down seven numbers, circling two of the numbers, then stating ' $3 = 3$ ' will not score full marks. Students needed to show the seven numbers, but then they had to state; ' $\max(\text{row}) = 3$ ' and ' $\min(\text{col}) = 3$ ', followed by ' $\text{as } \max(\text{row}) = \min(\text{col}), \text{ there is a stable solution}$ '. The play-safe strategies were well-answered, but there were very few correct answers for the saddle point.

Q3.

- (a) (i) Explanations rarely scored full marks. In order to show that the game has a stable solution, it was expected that the maximum values in the columns would be indicated before finding the minimum value of these maxima. The maximum of the row minima also needed to be found. Some statement should then have been made indicating that these two values were equal and hence that the game has a stable solution. Quite a few students felt that all they needed to do was to draw arrows pointing to -3 when more detail was required. A few chose to solve the problem using an argument based on dominance, but the explanations were rarely adequate to earn more than a single method mark.
- (ii) It was encouraging to see many students correctly identify the play-safe strategy for each player.
- (iii) Only the better students stated the correct value of the game for **Bill**, despite the bold type; as you might expect weaker students gave the value as -3 , and others gave the value as 0, possibly confused by the term "zero-sum game".
- (b) *A large number of students were not prepared for this part of the question.*
 - (i) A large number deleted the first column and scored only a couple of method marks for the rest of this question. Some who realised that the computer should not play C_2 spoiled their answers by writing such things as "each of the other strategies is better".
 - (ii) The better students managed to use the values in the table correctly to find correct expressions for the expected gains for Roza for the two remaining strategies. A few ignored the dominance from part(b)(i) and attempted to solve

the problem graphically using three lines; some successfully, others not.

- (iii) Some poor algebra, usually when removing brackets, prevented some from finding the correct value of p , but most marks were lost for having the incorrect pair of simultaneous equations.
- (iv) Those who scored full marks in parts (b)(ii) and (iii) were usually able to obtain the correct value of the game for Roza.

Q4.

Some candidates argued correctly, in part (a), that row 2 dominated row 3 but then incorrectly that one column dominated another in an attempt to reduce the matrix to a 2 by 2. In order to show that the game had a stable solution, it was expected that the minimum values in the rows and maximum values in the columns would be indicated before finding the maximum of the minima and the minimum of the maxima. Some statement should then have been made indicating that these two values were equal and hence that the game has a stable solution. The actual play-safe strategies for Ann and Bill were sometimes stated incorrectly even when the correct minimax and maximin values had been identified in the table.

The method of finding the optimal mixed strategy in part (b) seemed to have been clearly understood by most candidates. Sometimes the graphs were inaccurate and a mark was lost.

Other good solutions were sometimes spoiled by not stating how often Russ should play each of the strategies R_1 and R_2 . Those who found the correct probabilities in part (b)(i) were usually able to find the value of the game in part (b)(ii), but some substituted into the wrong expression for expected gain.

Q5.

In part (a), in order to show that the game had a stable solution, it was expected that the minimum values in the rows and maximum values in the columns would be indicated before finding the maximum of the minima and the minimum of the maxima. Some statement should then have been made indicating that these two values were equal and hence the game has a stable solution.

Poor algebraic skills in part (b), together with the omission of brackets, caused many candidates to write down expressions which were often simplified incorrectly. Many failed to make progress because they failed to equate each of the expressions found in part (b)(i) to $8/3$; others obtained nonsensical answers such as $p = 4/3$ because of earlier errors in their equations. It was surprising how many took more than a page of working to solve these two simple linear equations.

Q6.

Part (a) The term “zero-sum” was generally understood, but candidates should have indicated that the sum of Rob’s winnings and Con’s winnings was equal to zero *for every pair of strategies*.

Part (b) In order to show that the game has a stable solution, it was expected that the minimum values in the rows and the maximum values in the columns would be indicated before finding the maximum of the minima and the minimum of the maxima. Some statement should then have been made indicating that these two values are equal and hence the game has a stable solution.

Part (c) Most candidates realised that R_1 dominated R_3 and hence Rob should never play R_3 .

Part (d) To find the optimal mixed strategy, when a letter such as p is introduced, there should be an indication that this is the probability that Rob is choosing R_1 , for example. Three expressions in p were often written down with no indication as to what they represented. Although three linear graphs were often drawn, many candidates chose the wrong pair to solve for the optimal strategy and seemed to be guessing rather than reasoning correctly which point to select. Those who obtained the correct optimal strategy were usually able to find the value of the game.

Q7.

In part (a), in order to show that the game has a stable solution, it was expected that the minimum values in the rows and maximum values in the columns would be indicated before finding the maximum of the minima and the minimum of the maxima. Some statement should then have been made indicating that these two values are equal and hence the game has a stable solution. To find the optimal mixed strategy in part (b), when a letter such as p is introduced, there should be an indication that this is the probability that Ros is choosing R_1 , for example. Three expressions in p were often written down with no indication as to what they represented. Although three linear graphs were often drawn, many candidates chose the wrong pair to solve for the optimal strategy and seemed to be guessing rather than reasoning correctly which point to select. Those who obtained the correct optimal strategy were usually able to find the value of the game.

Q8.

Again this question was well answered, with many complete solutions to a conceptually difficult question. This was the first time that a game theory question had been set in which the values were algebraic. Some candidates solved for p first, others for x , with equal success.

Q9.

This question discriminated well. Candidates who had a good understanding of game theory scored maximum marks but the majority of candidates could not correctly identify the saddle points. It was essential that candidates were able to analyse all the different aspects of game theory mentioned in the specification.